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An interpretable machine learning model integrating clinical and CT radiomic features for the detection of interstitial lung disease in idiopathic inflammatory myopathies

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Abstract

Background Interstitial lung disease (ILD) is a leading cause of morbidity and mortality in patients with idiopathic inflammatory myopathy (IIM). However, early and accurate identification remains a clinical challenge. We aimed to develop and validate an interpretable machine learning framework to predict ILD presence in IIM patients.

Methods A retrospective cohort of 115 IIM patients ($n = 115$, including 70 with ILD) was divided into training ($n = 81$) and test ($n = 34$) sets. 1,316 radiomic features were extracted from automated lung segmentations of high-resolution CT (HRCT) scans. To prevent data leakage, feature selection—including correlation filtering, mRMR, and LASSO regression—was performed strictly within the training set. Multiple algorithms were evaluated via five-fold cross-validation, with LightGBM selected as the optimal framework. Independent clinical predictors were identified via multivariable logistic regression and integrated with the Rad-score to construct a combined model, visualized using a nomogram.

Results The combined clinical–radiomic model demonstrated superior discriminative ability, achieving an AUC of 0.877 (95% CI: 0.792–0.962) in the training set and 0.898 (95% CI: 0.791–1.000) in the independent test set, significantly outperforming clinical-only and radiomics-only models. Calibration curves and Decision Curve Analysis (DCA) confirmed the model's high predictive accuracy and clinical net benefit. SHAP analysis identified key radiomic features (e.g., wavelet.HLL_glcM_MCC) contributing to the model. Serum IgG concentration and presence of cough were identified as independent clinical predictors.

Conclusion Our interpretable machine learning nomogram, integrating clinical risk factors and CT radiomics, enables accurate and non-invasive detection of ILD in IIM patients. This framework provides a standardized approach

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for early screening, supporting timely intervention and improved clinical management for patients with pulmonary involvement.

Keywords Idiopathic inflammatory myopathy, Interstitial lung disease, Machine learning, Radiomics, LightGBM, Nomogram

Introduction

Idiopathic inflammatory myopathy (IIM) represents a heterogeneous group of autoimmune disorders characterized primarily by muscle inflammation and progressive muscle weakness. Major clinical subtypes include dermatomyositis, polymyositis, inclusion body myositis, and anti-synthetase syndrome [1]. Beyond skeletal muscle involvement, IIM is frequently accompanied by extra-muscular manifestations, such as cutaneous lesions, arthritis, and, most critically, interstitial lung disease (ILD) [2–4]. ILD is one of the most severe complications, often progressing rapidly and contributing to poor prognosis. Early detection is challenging because of subclinical presentations, whereas late-stage fibrotic changes are associated with a high mortality risk [5]. Thus, there is an urgent need for reliable methods to promptly identify IIM patients at risk of developing ILD.

With the rapid advancement of machine learning in recent years, its value in clinical diagnostics and risk assessment has become increasingly evident. High-resolution computed tomography (HRCT) has long been considered the gold standard for ILD evaluation [6, 7]. However, reliance on visual interpretation alone can lead to considerable inter-observer variability, even among experienced radiologists, underscoring the need for more objective and standardized assessment tools [8]. Emerging evidence suggests that machine learning algorithms integrating imaging and serological data demonstrate promising predictive performance in phenotypic analyses of IIM [1]. Additionally, machine learning has been applied to detect ILD in high-risk populations, to estimate the extent of pulmonary fibrosis, and to correlate radiological abnormalities with functional decline [9]. Despite these advances, early identification of ILD in IIM remains a major clinical challenge [2, 3]. Given the dismal prognosis associated with ILD, predictive models that integrate clinical features, validated biomarkers, and HRCT data are urgently needed [10].

In this context, machine learning offers an opportunity to enhance diagnostic accuracy, improve specificity, and facilitate personalized management strategies. By combining machine learning techniques with conventional diagnostic modalities, it may be possible to identify IIM patients at risk of ILD at an earlier stage, thereby reducing morbidity and mortality.

The present study aims to conduct a comprehensive analysis of a well-characterized IIM cohort to investigate demographic, clinical, and laboratory factors associated

with ILD. We seek to develop an effective non-invasive predictive model that integrates validated biomarkers with HRCT and clinical features to assess ILD risk in IIM patients. This work highlights not only the potential of machine learning in constructing clinically applicable radiomics models but also its broader role in advancing individualized care for patients with IIM.

Materials and methods

Study design and population

This retrospective cohort study included 115 patients diagnosed with IIM at the First Affiliated Hospital of Henan Medical University, China. The diagnosis of IIM was established according to the Bohan and Peter criteria [11] and the 2017 EULAR/ACR classification criteria [9]. The confirmation of ILD followed a rigorous standardized protocol to ensure diagnostic accuracy. First, HRCT scans were independently evaluated by two senior radiologists. To minimize diagnostic ambiguity, patients exhibiting confounding imaging features—such as ground-glass opacities (GGO) of unknown etiology, isolated pulmonary cysts, or gravity-dependent opacities—were excluded from the ILD cohort. Subsequently, the imaging findings were reviewed by an interstitial lung disease specialist in conjunction with clinical data. Finally, the definitive diagnosis of ILD was established through a multidisciplinary team (MDT) discussion involving experts from the departments of Rheumatology, Pathology, Radiology, and Pulmonology.

Eligible participants were adults (>18 years) with confirmed IIM who underwent HRCT for ILD evaluation. Exclusion criteria were: (1) alternative causes of myopathy (e.g., myasthenia gravis, muscular dystrophy, infections, drugs, or endocrine disorders); (2) concomitant pulmonary diseases confounding ILD diagnosis (e.g., tuberculosis, or drug-induced lung injury); and (3) incomplete clinical or imaging data.

Demographic and clinical data were collected, including age, sex, BMI, presence of cancer, and pectoral muscle parameters. Clinical symptoms such as cough, dyspnea, hemoptysis, rash, muscle weakness, facial edema, mechanic's hands, and arthralgia were recorded. Laboratory indicators included inflammatory markers (ESR, CRP); liver function indices (ALT, AST, ALB); hematologic parameters; immune indicators (IgA, IgG, IgM, C3, C4); muscle enzymes (LDH, CK, CK-MB); and myositis-specific or -related antibodies (RO52, IIF, SSB, SAE, AHA, SRP-IgG, U1-snRNP, SP100, CENPB, Jo-1,

PL-12, EJ, NXP2, HMGCR, M2, KU, MDA5, TIF1 γ , SCL7, and SmD1). All patients underwent HRCT for ILD assessment.

Whole lung CT image segmentation

Automatic lung segmentation was performed on HRCT images using the open-source U-Net-based deep learning model R231 (<https://github.com/JoHof/LUNGMA5K>). This model has been extensively validated across heterogeneous datasets, demonstrating robust performance across various scanners and acquisition protocols [12]. Regions of interest (ROIs) for the left and right lungs were automatically generated and subsequently subjected to rigorous visual inspection by two experienced observers to ensure anatomical plausibility and complete lung coverage (Fig. 1). Cases with segmentation errors were excluded. Given the established accuracy of the R231 model, no additional manual segmentation-based validation was performed.

Radiomics feature extraction and image preprocessing

Radiomic features were extracted using PyRadiomics (version 3.1.0; Python 3.9) with SimpleITK (version 2.5.2) as the backend [13]. To ensure reproducibility and adherence to the Image Biomarker Standardization Initiative (IBSI) guidelines, all HRCT images were resampled to an isotropic voxel spacing of $1.0 \times 1.0 \times 1.0$ mm³ using linear interpolation. Image discretization was performed with a fixed bin width of 25 HU, and no additional intensity normalization was applied.

A total of 1,316 quantitative features were extracted from the original images and 13 filtered derivatives,

including wavelet-transformed and Laplacian of Gaussian (LoG) filtered images. Specifically, LoG filtering was implemented with sigma values ranging from 1 to 5 to capture image features at multiple spatial scales, and three wavelet-decomposed images (HLL, LHL, and LLH) were generated to extract texture features reflecting different frequency and directional components. This feature set encompassed three categories: first-order statistics, shape descriptors, and textural matrices, with the latter including the gray-level co-occurrence matrix (GLCM), gray-level run-length matrix (GLRLM), gray-level size zone matrix (GLSZM), and neighborhood gray-level difference matrix (NGLDM).

Feature selection and rad-score construction

To ensure objective evaluation and prevent data leakage, the dataset was first partitioned into training and testing cohorts at a 7:3 ratio using stratified sampling, with all subsequent preprocessing and selection steps performed strictly within the training cohort. Z-score normalization was applied to the training set, and the resulting scaling parameters were subsequently projected onto the test set to maintain independent validation. Feature selection followed a nested hierarchy, beginning with redundancy reduction where the choice between Pearson and Spearman correlation was determined by the Shapiro-Wilk normality test ($p > 0.05$ for Pearson); for feature pairs with an absolute correlation coefficient > 0.90 , the feature with the lower correlation to ILD status was excluded. The remaining features were ranked using the minimum redundancy maximum relevance (mRMR) algorithm, followed by LASSO regression with five-fold

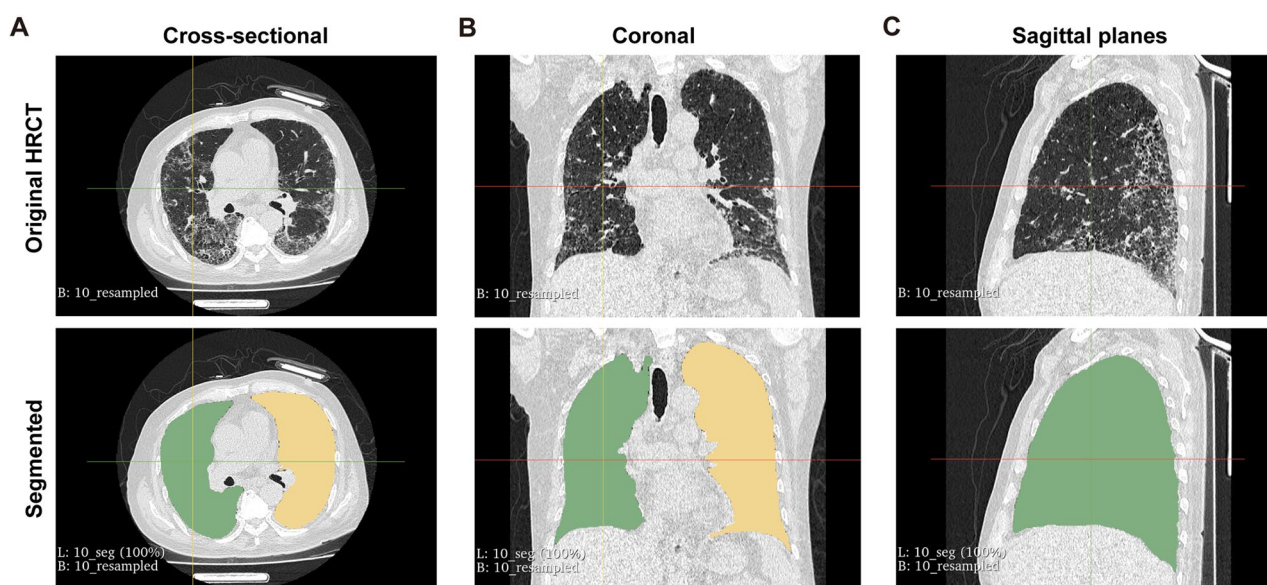


Fig. 1 Representative original chest HRCT images and corresponding segmented lung regions in patients with IIM and IIM-associated ILD. Panels (A–C) show cross-sectional (A), coronal (B), and sagittal (C) views from one patient

cross-validation. Finally, A Rad-score was then calculated for each patient using the formula:

$$[Rad-score = \sum_{i=1}^n \beta_i \times X_i]$$

where X_i represents the standardized value of the i -th selected radiomics feature, and β_i denotes the corresponding coefficient derived from the LASSO model. The same feature set and coefficients obtained from the training data were subsequently applied to the test data to calculate Rad-score.

Machine learning model construction and evaluation

Based on the identified feature subset, six machine learning algorithms were implemented: Random Forest (RF), Decision Tree (DT), Extreme Gradient Boosting (XGBoost), K-Nearest Neighbor (KNN), Support Vector Machine (SVM), and Light Gradient Boosting Machine (LightGBM). To optimize hyperparameters and ensure stability, five-fold cross-validation was applied within the training set. AUC, accuracy, sensitivity, and specificity were used to evaluate and compare diagnostic efficacy.

Development of the machine learning-based combined model

A joint clinical–radiomic model was developed using the LightGBM algorithm to evaluate the predictive performance of the integrated feature set. LightGBM was specifically selected for its superior classification power and its ability to capture complex, non-linear relationships between high-dimensional radiomic features and clinical information. Model performance was rigorously evaluated using receiver operating characteristic (ROC) curve analysis in both training and test cohorts. While this algorithmic approach was utilized to establish the optimal diagnostic efficacy of the integrated dataset, its “black-box” nature necessitates an additional interpretable tool for clinical practice.

Clinical variable selection and nomogram construction

To translate the predictive potential of the combined features into an accessible clinical tool, the clinical variables were analyzed using the same 7:3 training/testing split to ensure comparability. Within the training set, univariate logistic regression was first performed on clinical variables to screen for potential predictors ($p < 0.05$). To prioritize interpretability, a multivariable logistic regression model (using the rms R package) was then employed to identify independent predictors of ILD. Odds ratios (ORs) and 95% confidence intervals (CIs) were calculated for each variable. Finally, a combined nomogram

was constructed by integrating the Rad-score with these independent clinical factors. This traditional statistical approach serves as a clinical expression tool for the combined model, providing an interpretable and visualized system that avoids the “black box” complexity of machine learning while facilitating individualized risk estimation.

Model validation and comparison

Decision curve analysis was applied to assess the clinical utility of the models by quantifying net benefits across a range of risk thresholds. The radiomics, clinical, combined, and null (featureless) models were compared to evaluate their potential impact on clinical decision-making.

Statistical analysis

All statistical analyses were performed using R software (v4.2.2). Continuous variables were first evaluated for normality using the Shapiro–Wilk test. Variables that conformed to a normal distribution were expressed as mean $\pm$ standard deviation, and intergroup comparisons were performed using the independent samples t-test; variables that did not conform to a normal distribution were expressed as median (interquartile range), and intergroup comparisons were performed using the Wilcoxon rank sum test. Categorical variables were expressed as the number of cases (percentage), and the chi-square test was used for comparison between groups. When the theoretical frequency was less than 5, Fisher's exact test was used. All statistical tests were two-sided, and $p < 0.05$ was considered statistically significant.

Result

Case screening

A total of 241 patients diagnosed with IIM were initially identified at the First Affiliated Hospital of Henan Medical University from August 2018 to July 2023. Among these, 50 patients were excluded due to uncertain diagnosis, 70 were excluded for lacking chest CT scans, and 6 were excluded because they were aged ≤ 18 years. The remaining 115 patients were included in the study and categorized into two groups: 45 patients (32 females, median age 48.67 years) with IIM without ILD, and 70 patients (45 females, median age 59.06 years) with IIM-ILD (Fig. 2).

Radiomics feature screening and Rad-score establishment

A total of 1,316 radiomic features were initially extracted from the HRCT images and normalized using Z-score transformation. Pearson or Spearman correlation analysis was first applied to remove highly correlated features, resulting in the exclusion of 1,110 features and leaving 206 features for subsequent analysis (Fig. 3A). Feature selection was then performed using the LASSO

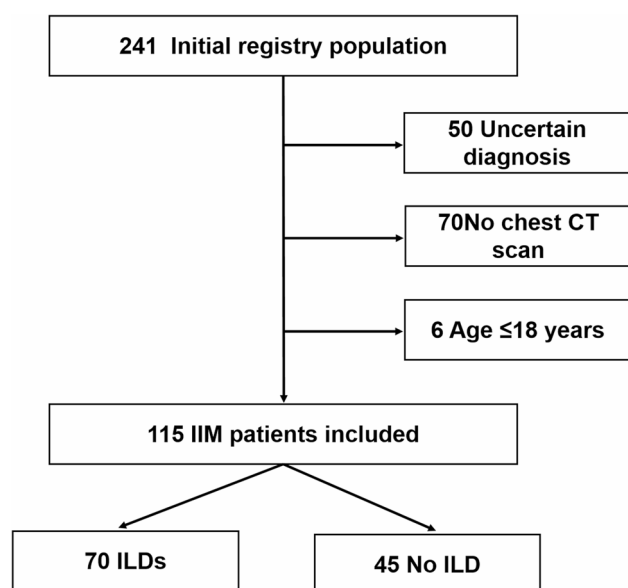


Fig. 2 Flow chart for patient inclusion and exclusion

regression with five-fold cross-validation. The optimal regularization parameter ($\lambda = 0.0293$) was determined, yielding 15 features with non-zero coefficients (Figs. 3B–D). The LASSO coefficient distribution (Fig. 3B) highlighted the most predictive features, with the vertical red line indicating the optimal λ that balanced model complexity and predictive accuracy. Stability of feature selection across cross-validation folds was demonstrated by the consistency of non-zero feature counts, as indicated in the cross-validation plot (Fig. 3C). The pairwise correlations among the selected features were visualized using a heatmap, where red and purple represented positive and negative correlations, respectively, with darker shades indicating stronger correlations (Fig. 3D). Finally, the coefficients of the 15 selected features were used to compute a Rad-score for each patient, summarized in a histogram (Fig. 3E). These features were linearly combined by weighting their coefficients, and the resulting Rad-score (Supplementary file 1) was subsequently applied in predictive modeling for ILD outcomes.

Machine learning model performance and optimization

To ensure robust model performance, hyperparameter tuning for all machine learning algorithms was conducted within the tidymodels framework in R, using a grid search strategy coupled with five-fold cross-validation strictly within the training set. The optimal hyperparameters were selected based on the highest average Area Under the Receiver Operating Characteristic Curve (AUC) across the validation folds. The test set remained independent and was only utilized for the final performance evaluation after the hyperparameters were finalized (Supplementary file 2).

The discriminative performance of the six optimized models is summarized (Table 1; Fig. 4). The Decision Tree (DT) model achieved a training AUC of 0.922 and a test AUC of 0.784 (Fig. 4A). The Random Forest (RF) model exhibited a training AUC of 0.963 and a test AUC of 0.828 (Fig. 4B). The XGBoost model demonstrated a training AUC of 0.933 and a test AUC of 0.814 (Fig. 4C). The Support Vector Machine (SVM) model achieved a training AUC of 0.975 and a test AUC of 0.837 (Fig. 4D). The K-Nearest Neighbors (KNN) model showed a training AUC of 0.966 and a test AUC of 0.873 (Fig. 4E). Finally, the LightGBM model exhibited a training AUC of 0.963 and a test AUC of 0.852 (Fig. 4F).

Among the evaluated algorithms, the LightGBM model demonstrated a robust balance between predictive power and stability. Although the KNN model exhibited a marginally higher point estimate for the test AUC, DeLong's test confirmed no statistically significant difference in discriminative performance between LightGBM and KNN ($p > 0.05$; Supplementary file 3). Notably, in the test set, the LightGBM model surpassed KNN in other clinical metrics, including accuracy, specificity, and precision.

The selection of LightGBM as our final predictive framework was further supported by its superior generalization capability. Unlike distance-based methods such as KNN, which are often sensitive to feature redundancy and noise in high-dimensional datasets, LightGBM—a tree-based ensemble method—demonstrated greater resilience to multicollinearity and more consistent performance across internal validation. This robustness, combined with its high efficiency and scalability, identifies LightGBM as the most suitable model for potential clinical deployment.

Visualizing the importance of lightGBM models for imaging features

Shapley Additive Explanations (SHAP) were employed to interpret the predictions of the LightGBM radiomics model, quantifying both the magnitude and direction of each feature's contribution to the predicted outcome. SHAP analysis identified key features, including wavelet. HLL_glcM_MCC and original_glszm_LargeAreaHigh-GrayLevelEmphasis, which collectively influenced the final prediction through positive and negative adjustments (Figs. 5A–C). These findings not only confirm the interpretability of the LightGBM model but also provide a basis for subsequent selection of clinically relevant imaging features. To enhance the interpretability of the model, we constructed a Rad-score based on the selected radiomics features, which was used for the subsequent construction of the combined model and was regarded as a single comprehensive radiomics prediction indicator in the subsequent analysis.

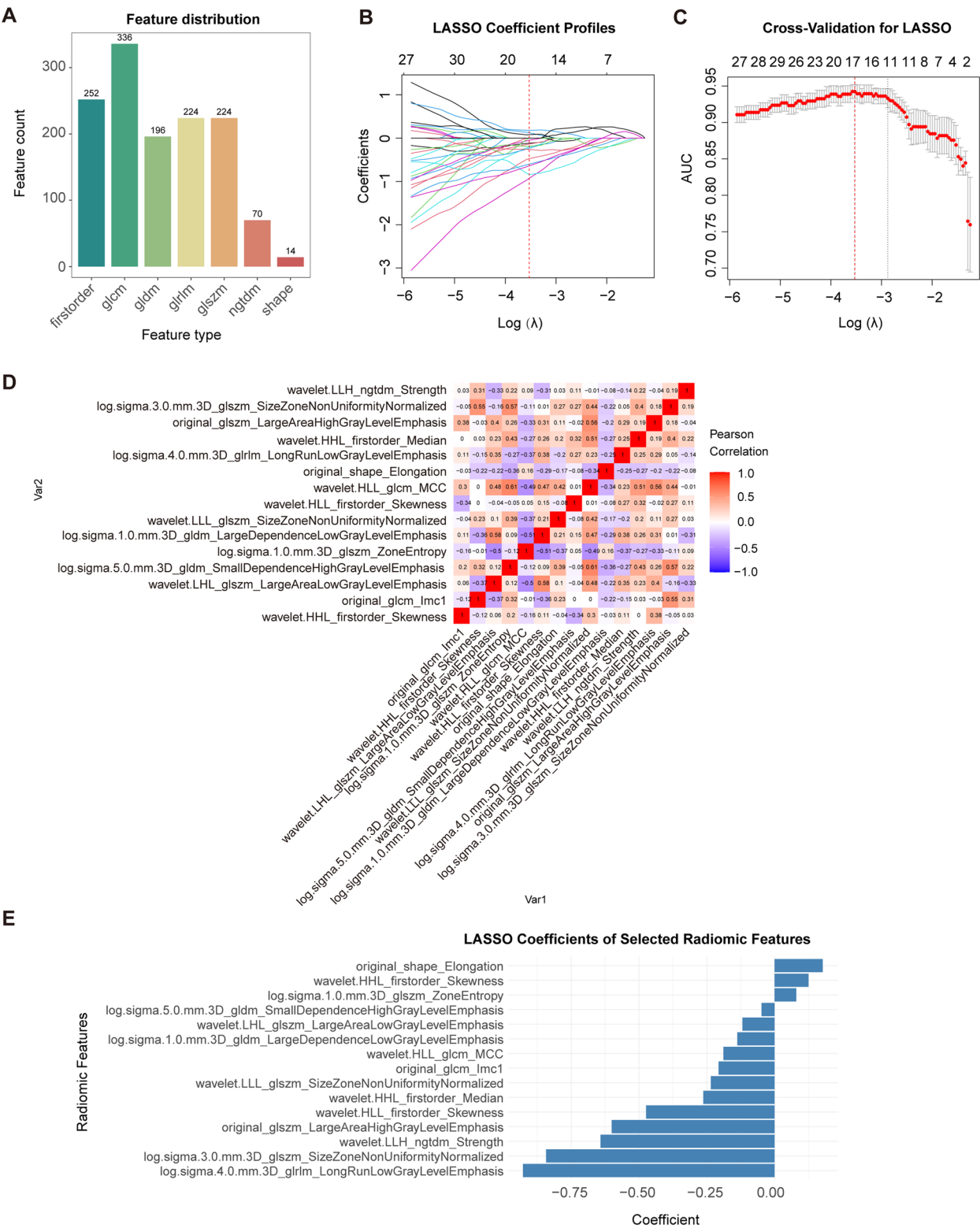


Fig. 3 LASSO coefficient plots of imaging features. **A** Feature classification statistics. **B** Distribution of LASSO coefficients for 206 imaging features. Five-fold cross-validation was performed, and a vertical line marks the logarithmic (λ) value where the optimal λ (0.02933494) selected 15 imaging features; the upper X-axis indicates the number of nonzero coefficient features. **C** The black and red vertical lines denote the λ values selected from five-fold cross-validation in panel B, with the upper X-axis again showing the count of nonzero features. **D** Heatmap showing correlations among different imaging features, where red represents positive and purple negative correlations, and darker colors indicate stronger relationships. **E** Radcore histogram of the selected radiomic features; the Y-axis lists the 15 selected features, and the X-axis shows their corresponding coefficients

Table 1 Performance Comparison of Six Machine Learning Algorithms

DT train	Accuracy	Sensitivity	Specificity	Precision	f1_score	AUC
	0.864	0.958	0.727	0.836	0.893	0.922
test	0.706	0.818	0.500	0.750	0.783	0.784
RF train	0.901	0.896	0.909	0.935	0.915	0.963
test	0.706	0.682	0.750	0.833	0.750	0.828
xgboost train	0.864	0.833	0.909	0.930	0.879	0.933
test	0.706	0.682	0.750	0.833	0.750	0.814
svm train	0.926	0.896	0.970	0.977	0.935	0.975
test	0.706	0.682	0.750	0.833	0.750	0.837
knn train	0.901	0.896	0.909	0.935	0.915	0.966
test	0.706	0.682	0.750	0.833	0.750	0.873
lightgbm train	0.889	0.833	0.970	0.976	0.899	0.963
test	0.735	0.682	0.833	0.882	0.769	0.852

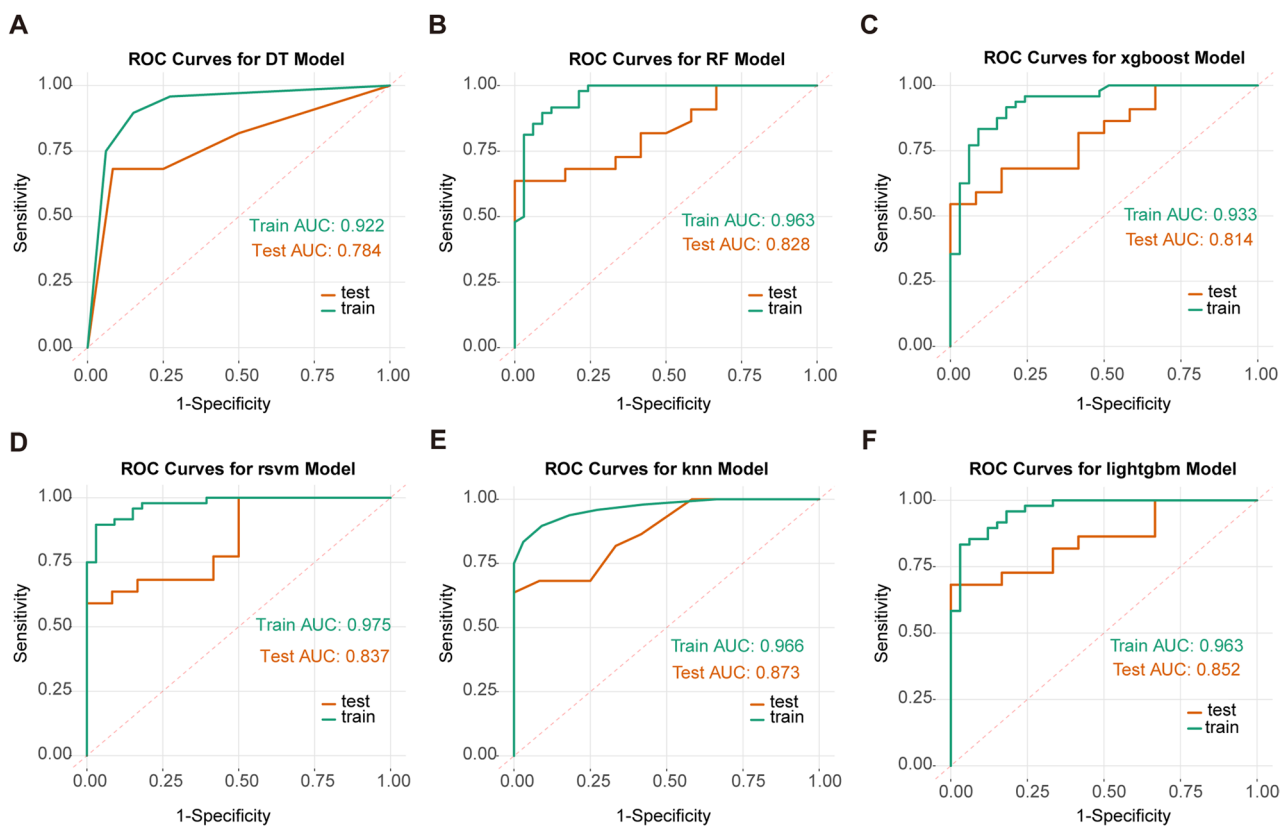


Fig. 4 ROC curves of radiomics models constructed using different machine learning algorithms. **A** Decision Tree (DT), **B** Random Forest (RF), **C** XGBoost, **D** Support Vector Machine (SVM), **E** K-Nearest Neighbors (KNN), and **F** LightGBM. The horizontal axis indicates specificity, and the vertical axis indicates sensitivity. The green curve represents the training set, whereas the orange curve represents the test set

Clinical feature screening

Analysis of the patient records revealed several significant differences between the IIM-ILD and IIM groups. Patients in the IIM-ILD group were significantly older than those in the IIM group, while no significant difference in gender distribution was observed. Regarding clinical manifestations, respiratory symptoms were significantly more frequent in the IIM-ILD group, characterized by a higher incidence of cough and dyspnea. In contrast, muscle weakness was significantly more

prevalent in the IIM group without ILD. Other systemic and musculoskeletal symptoms, including fever, dizziness, chest pain, hemoptysis, rash, facial edema, mechanic’s hands, and arthralgia, did not differ significantly between the two groups (Table 2).

Laboratory analyses demonstrated significantly higher inflammatory markers (ESR and CRP) and elevated IgG levels in the IIM-ILD group. Conversely, the IIM group exhibited significantly higher levels of ALT and ALB. Other laboratory parameters, including hepatic and

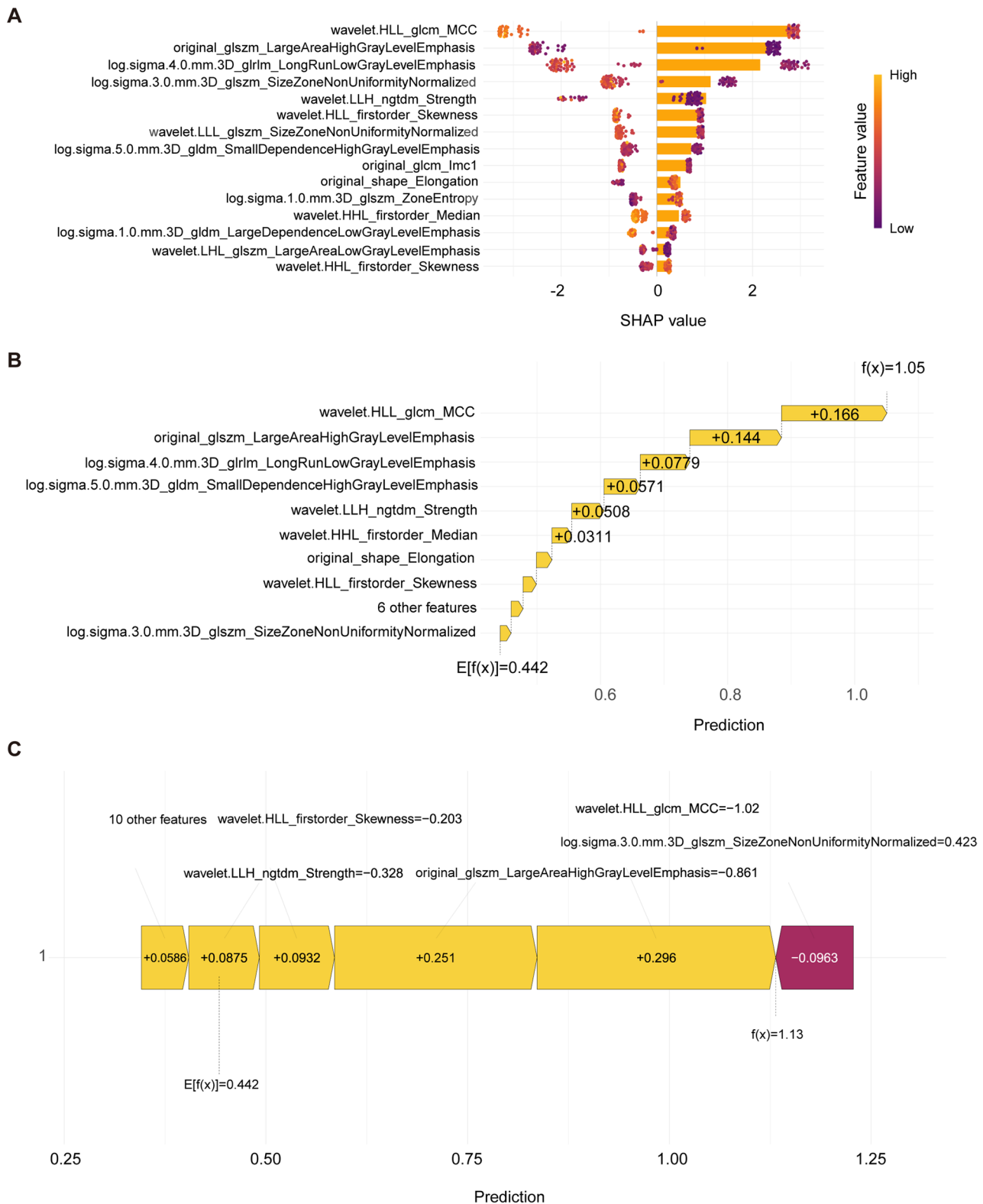


Fig. 5 Visualization of imaging feature importance in the LightGBM model. **A** SHAP summary plot illustrating the contribution of each imaging feature to model predictions. The X-axis represents SHAP values, where positive values indicate positive contributions and negative values indicate inhibitory effects. The Y-axis lists feature names, with darker-colored points representing features with higher impact on model output. **B** SHAP waterfall plot of a high-prediction case, showing how individual features contribute to the final model output. Yellow bars represent positive contributions, and longer bars correspond to features with greater predictive influence. **C** SHAP waterfall plot of a low-prediction case, where red bars indicate negative contributions, and the bar length reflects the magnitude of each feature's effect on the prediction

Table 2 Comparison of Categorical Clinical Variables Between IIM and IIM-ILD Groups

Characteristic	(n = 115)	0 ¹ (n = 45)	1 ¹ (n = 70)	p-value ²
Gender				0.448
Male	38 (33%)	13 (29%)	25 (36%)	
Female	77 (67%)	32 (71%)	45 (64%)	
Cancer	17 (15%)	6 (13%)	11 (16%)	0.726
Autoantibodies				
RO52	51 (44%)	20 (44%)	31 (44%)	0.987
IIF	99 (86%)	39 (87%)	60 (86%)	0.885
SSB	16 (14%)	5 (11%)	11 (16%)	0.486
SAE	9 (7.8%)	3 (6.7%)	6 (8.6%)	> 0.999
AHA	3 (2.6%)	0 (0%)	3 (4.3%)	0.279
SRPIgG	5 (4.3%)	3 (6.7%)	2 (2.9%)	0.378
U1snRNP	5 (4.3%)	0 (0%)	5 (7.1%)	0.155
SP100	1 (0.9%)	0 (0%)	1 (1.4%)	> 0.999
CENPB	1 (0.9%)	0 (0%)	1 (1.4%)	> 0.999
Jo1	16 (14%)	3 (6.7%)	13 (19%)	0.072
PL12	9 (7.8%)	1 (2.2%)	8 (11%)	0.088
EJ	5 (4.3%)	1 (2.2%)	4 (5.7%)	0.647
NXP2	5 (4.3%)	2 (4.4%)	3 (4.3%)	> 0.999
HMGCR	2 (1.7%)	1 (2.2%)	1 (1.4%)	> 0.999
M2	1 (0.9%)	1 (2.2%)	0 (0%)	0.391
KU	2 (1.7%)	1 (2.2%)	1 (1.4%)	> 0.999
MDA5	11 (9.6%)	4 (8.9%)	7 (10%)	> 0.999
TIF1γ	6 (5.2%)	3 (6.7%)	3 (4.3%)	0.677
SCL7	2 (1.7%)	0 (0%)	2 (2.9%)	0.519
SmD1	3 (2.6%)	0 (0%)	3 (4.3%)	0.279
Clinical manifestations				
Cough	45 (39%)	5 (11%)	40 (57%)	< 0.001
Dyspnea	50 (43%)	7 (16%)	43 (61%)	< 0.001
Hemoptysis	2 (1.7%)	1 (2.2%)	1 (1.4%)	> 0.999
Chestpain	1 (0.9%)	1 (2.2%)	0 (0%)	0.391
Rash	63 (55%)	29 (64%)	34 (49%)	0.095
Muscle weakness	60 (52%)	29 (64%)	31 (44%)	0.035
Mechanic's hands	30 (26%)	11 (24%)	19 (27%)	0.748
Faceedema	4 (3.5%)	3 (6.7%)	1 (1.4%)	0.298
Arthralgia	7 (6.1%)	2 (4.4%)	5 (7.1%)	0.703
Fever	24 (21%)	6 (13%)	18 (26%)	0.111
Dizzy	5 (4.3%)	2 (4.4%)	3 (4.3%)	> 0.999

¹n (%)²Pearson's Chi-squared test; Fisher's exact test³Boldface values indicate statistical significance ($p < 0.05$)

muscle enzymes, peripheral blood indices, IgA, IgM, and complement components (C3, C4), showed no significant differences between the groups. Furthermore, the distribution of myositis-related antibodies was statistically similar across both cohorts, indicating comparable serological profiles. Overall, older age, respiratory symptoms (cough and dyspnea), muscle weakness, elevated inflammatory and immunological markers (ESR, CRP, and IgG), and lower levels of ALT and ALB were the primary clinical features distinguishing IIM-ILD from isolated IIM patients (Table 3).

Independent predictors of IIM-ILD

The aforementioned clinical variables were included in the univariate regression analysis, and variables with a $p < 0.05$ in the univariate regression analysis were further incorporated into the multivariate regression analysis (Table 4). In univariate analysis, age, ESR, ALT, AST, IgA, IgG, cough, dyspnea, and muscle weakness were significantly associated with IIM-ILD. After incorporating these variables with statistical significance into the multivariate logistic regression model, it was determined that IgG and cough are independent predictors of ILD in IIM patients. Specifically, IgG and cough remained significant, Other variables including age, ESR, ALT, AST, IgA, dyspnea, and muscle weakness lost significance in the multivariate model, indicating that their effects were confounded by the selected predictors. These results indicate that elevated IgG, presence of cough are independent risk factors for ILD development among IIM patients.

Visual combined model and its performance evaluation

To facilitate clinical application, an interpretable nomogram was constructed as a clinical expression tool by integrating the Rad-score with two independent clinical predictors using a multivariable logistic regression framework (Fig. 6A). This tool enables clinicians to estimate the individual probability of IIM-associated ILD by summing the points assigned to each variable, thereby supporting timely clinical evaluation and intervention. The agreement between the predicted probabilities and actual ILD incidence was evaluated using calibration curves. In the training set, 1000 bootstrap resamples demonstrated that the bias-corrected calibration curve closely aligned with the ideal diagonal line, yielding a mean absolute error of 0.02 (Fig. 6B). In the independent test set, both non-parametric and logistic regression calibration analyses showed consistent performance (Fig. 6C). Quantitative metrics in the test set—including a Brier score of 0.048, an intercept of -0.249, and a slope of 1.207—confirmed that the model maintained reliable predictive accuracy with minimal systematic bias.

The clinical utility of the integrated framework was further assessed via Decision Curve Analysis (DCA). Across a wide range of risk thresholds (0.2–0.8), the combined model (green curve) provided a higher net benefit than both the clinical-only (blue curve) and radiomics-only (red curve) models in both the training (Fig. 6D) and test (Fig. 6E) cohorts, underscoring its practical value in clinical decision-making.

Finally, ROC analysis was utilized to compare the discriminative performance across models. In the training set, the combined model achieved an AUC of 0.877 (95% CI: 0.792–0.962), significantly outperforming single-modality approaches (Fig. 6F). This superior performance was sustained in the test set, where the combined model

Table 3 Comparison of Continuous Clinical Variables Between IIM and IIM-ILD Groups

Characteristic	Overall ¹ (n = 115)	0 ¹ (n = 45)	1 ¹ (n = 70)	p-value ²
Age (years)	55 ± 14	49 ± 16	59 ± 11	<0.001
Height(cm)	164 ± 8	165 ± 8	163 ± 8	0.135
Weight(Kg)	62 ± 12	64 ± 13	61 ± 11	0.176
BMI(kg/m ²)	22.9 (20.8, 25.4)	23.0 (20.9, 25.4)	22.7 (20.8, 25.3)	0.488
Pectoralmusclesright(mm)	14.7 (11.3, 19.1)	14.6 (11.1, 17.9)	14.9 (12.3, 19.2)	0.547
Pectoralmuscleleft(mm)	14.9 ± 4.9	15.5 ± 5.2	14.5 ± 4.7	0.530
Totalpectoralarea(cm ²)	29 (23, 37)	28 (23, 38)	29 (23, 36)	0.975
FFM/PMA	41 (38, 51)	42 (38, 52)	41 (38, 50)	0.374
FFMI(kg/m ²)	16.21 ± 1.99	16.26 ± 2.16	16.18 ± 1.89	0.886
Laboratory indicators				
WBC(×10 ⁹ /L)	6.9 (5.3, 9.9)	6.8 (5.1, 9.9)	6.9 (5.5, 9.9)	0.547
NEU(×10 ⁹ /L)	5.33 (3.59, 7.81)	4.88 (3.49, 7.87)	5.37 (4.06, 7.55)	0.449
LEU(×10 ⁹ /L)	1.25 (0.77, 1.66)	1.22 (0.73, 1.56)	1.26 (0.79, 1.78)	0.669
MON(×10 ⁹ /L)	0.34 (0.24, 0.43)	0.38 (0.27, 0.44)	0.32 (0.24, 0.43)	0.188
PLT(×10 ⁹ /L)	231 (192, 297)	242 (202, 311)	228 (192, 289)	0.331
HB(g/L)	123 ± 16	121 ± 18	124 ± 15	0.368
ESR(mm/h)	18 (9, 33)	11 (6, 18)	27 (13, 35)	<0.001
CRP(mg/L)	3 (1, 10)	2 (1, 6)	4 (1, 11)	0.035
LDH(U/L)	294 (209, 443)	303 (214, 396)	277 (206, 453)	0.705
CK(U/L)	114 (45, 528)	148 (44, 734)	108 (46, 450)	0.768
CKMB(ng/mL)	9 (2, 26)	10 (2, 26)	8 (2, 23)	0.783
ALT(U/L)	33 (20, 68)	44 (27, 78)	31 (16, 50)	0.005
AST(U/L)	37 (21, 82)	43 (22, 113)	33 (21, 69)	0.059
ALB(g/L)	37.9 (34.2, 40.9)	39.4 (36.1, 42.0)	36.5 (33.6, 40.0)	0.009
C3(g/L)	1.01 (0.88, 1.21)	0.98 (0.85, 1.22)	1.03 (0.89, 1.20)	0.638
C4(g/L)	0.23 (0.17, 0.28)	0.22 (0.17, 0.26)	0.23 (0.18, 0.28)	0.482
IgA(g/L)	2.49 (1.76, 3.21)	2.20 (1.67, 2.90)	2.55 (1.89, 3.60)	0.053
IgG(g/L)	12.2 (9.7, 15.9)	10.5 (9.4, 12.9)	12.8 (10.8, 17.0)	0.001
IgM(g/L)	1.26 (0.77, 1.74)	1.26 (0.75, 1.75)	1.24 (0.81, 1.69)	0.848

¹Mean ± SD; Median (Q1, Q3)²Wilcoxon rank sum test**Table 4** Univariate and multivariate regression analysis table

Variable	Univariable analysis		Multivariable analysis	
	OR(95%CI)	p-value	OR(95%CI)	p-value
Age	1.060(1.027–1.094)	<0.001	1.042(0.998–1.087)	0.061
ESR	1.032(1.008–1.056)	0.009	1.020(0.984–1.058)	0.283
ALT	0.991(0.984–0.998)	0.010	1.002(0.992–1.013)	0.649
AST	0.993(0.986–0.999)	0.022	0.992(0.981–1.004)	0.199
IgA	1.503(1.042–2.167)	0.029	1.020(0.986–1.061)	0.466
IgG	1.181(1.063–1.312)	0.002	1.229(1.066–1.416)	0.004
cough	10.667(3.758–30.277)	<0.001	5.125(1.397–18.802)	0.014
dyspnea	8.646(3.381–22.107)	<0.001	2.978(0.864–10.261)	0.084
muscle weakness	0.439(0.203–0.948)	0.036	1.123(0.371–3.399)	0.837

reached an AUC of 0.898 (95% CI: 0.791–1.000) (Fig. 6G), indicating robust discriminative ability and favorable generalization without evidence of overfitting. Taken together, by integrating clinical risk factors with quantitative imaging phenotypes, the combined model exhibited superior performance in terms of discrimination, calibration, and clinical utility, suggesting its significant potential for clinical translation.

Discussion

In this single-center retrospective study of 115 IIM patients, we developed a LightGBM-based machine learning model integrating clinical variables and quantitative HRCT radiomics to predict coexistent ILD. Our analysis revealed that elevated serum IgG levels, and the presence of cough were independent predictors of ILD in patients with IIM. While the radiomics-only model demonstrated moderate discriminatory ability, the integrative clinical–radiomics model achieved markedly superior performance, evidenced by a higher AUC and greater net clinical benefit on decision curve analysis compared with

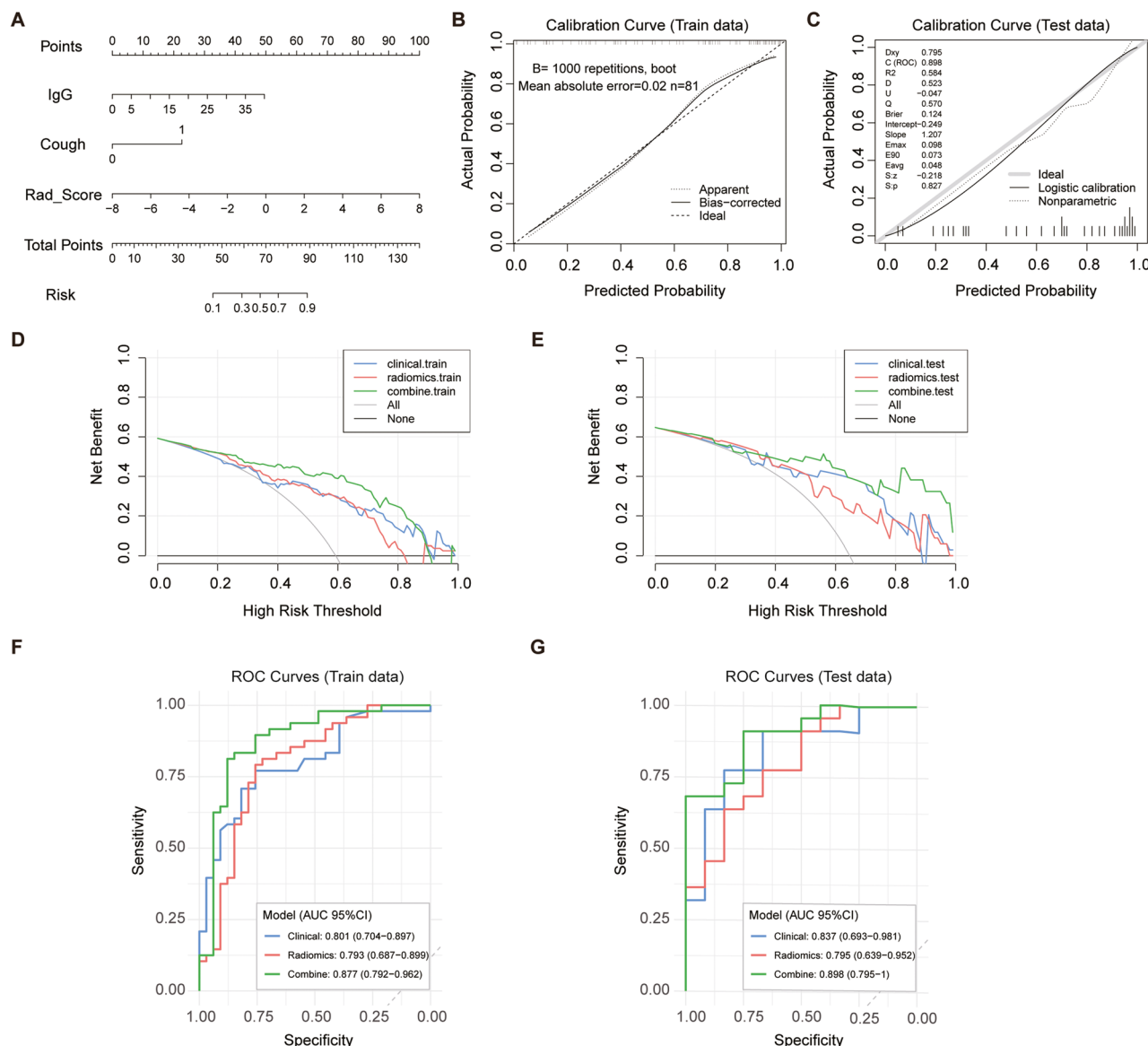


Fig. 6 Development and evaluation of the combined clinical-radiomic model. **A** Nomogram integrating independent clinical risk factors and the Rad-score for individualized probability estimation of IIM-associated ILD. **B, C** Calibration curves for the training (B, $n=81$) and test (C, $n=34$) sets. The dashed diagonal represents ideal calibration; solid lines indicate bias-corrected performance via 1000 bootstrap resamples (mean absolute error=0.02 in the training set). **D, E** Decision curve analysis (DCA) in the training and test sets. The combined model (green) consistently yields higher clinical net benefit than the radiomics-only (red) and clinical-only (blue) models across a wide range of risk thresholds. **F, G** ROC curves comparing the discriminative performance of the three models in the training (F) and test (G) sets, highlighting the superior and robust AUC of the integrated framework

models based on either domain alone. These key findings suggest that, while traditional clinical risk factors remain important, incorporating imaging-derived quantitative features can meaningfully improve ILD risk stratification in IIM.

ILD is a common and serious complication in IIM. In large cohorts, nearly half of IIM patients have some form of ILD [14, 15], and ILD is a major driver of morbidity and mortality in myositis [16, 17]. For example, some series report that ILD accounts for up to 50% of the excess mortality seen in IIM cohorts [14, 18]. Consistent

with these reports, nearly half of our cohort had ILD. Previous studies have identified various clinical and serological risk factors for ILD in myositis. Notably, certain myositis-specific autoantibodies—including anti-Ro-52, anti-PM/Scl [18], anti-aminoacyl tRNA synthetase [19], and anti-MDA5 [20]—have been significantly associated with ILD development. Although not all myositis autoantibodies were assessed, the observation that higher serum IgG levels independently predicted ILD risk is particularly noteworthy, as IgG reflects systemic immune activation. IgG may reflect underlying immunologic or

inflammatory burden in ILD, somewhat analogous to prior reports of elevated IgM or other markers in myositis-ILD [15]. To our knowledge, the association of serum IgG with ILD risk in IIM has not been widely reported, and it may warrant further study. Notably, we did not find any autoantibody such as anti-Ro52 or anti-MDA5 to emerge in multivariable analysis; the differences in identified predictors (IgG and symptom of cough, etc.) may relate to our cohort characteristics, local practice patterns, or statistical model differences.

The identification of cough as an independent predictor of IIM-ILD is biologically plausible but seldom emphasized in predictive models. Cough is one of the cardinal presenting symptoms of ILD (along with dyspnea), reflecting pulmonary involvement. Generally, in fibrotic ILD, chronic cough is both common and associated with worse patient-reported outcomes [21]. That cough entered our model suggests that IIM patients reporting cough, even in the absence of overt dyspnea, should raise concern for subclinical lung fibrosis. This symptom-based predictor likely captures aspects of lung involvement not reflected in laboratory values, complementing traditional risk factors like autoantibody status.

In our study, an intriguing finding was that muscle weakness was more prevalent in the non-ILD group and exhibited a negative association with ILD in the univariate analysis ($OR=0.439$, $p=0.036$). This observation, while seemingly counterintuitive, is consistent with the recognized clinical heterogeneity of IIM. Specifically, it aligns with the clinical phenotype of Clinically Amyopathic Dermatomyositis (CADM). Patients with CADM typically present with characteristic skin rashes but minimal to no objective muscle weakness; however, they carry a significantly higher risk of developing rapidly progressive ILD (RP-ILD). In our cohort, the ILD group likely encompassed a higher proportion of patients with such “amyopathic” or “hypomyopathic” phenotypes, where the systemic inflammatory response primarily targets the pulmonary parenchyma rather than skeletal muscle. Therefore, the absence of muscle weakness in certain IIM subsets should not be interpreted as a sign of lower systemic severity, but rather as a potential indicator of a lung-dominant phenotype that necessitates vigilant pulmonary screening.

Furthermore, our analysis revealed that serum albumin (ALB) levels were significantly lower in the ILD group compared to those without ILD. As a protein synthesized primarily by the liver, ALB levels are modulated by systemic inflammation, nutritional status, and hepatic function. In IIM-ILD, the underlying pathophysiological mechanism likely involves a potent systemic inflammatory response, where pro-inflammatory cytokines can competitively inhibit ALB synthesis and accelerate its consumption, effectively acting as a “negative

acute-phase reactant” [22]. While previous studies in Chinese cohorts have identified low serum albumin as an independent risk factor for both ILD development and mortality in IIM [15], ALB did not emerge as an independent predictor in our final multivariate model. This discrepancy may be attributed to our relatively small sample size or potential confounding effects from variables such as age and baseline nutritional status. Nonetheless, the clear trend observed in the univariate analysis and the marked clinical discrepancy between groups underscore its potential as a supplementary screening marker, particularly in resource-limited settings where advanced biomarkers may be unavailable. Future large-scale prospective studies are warranted to further clarify the independent role of ALB in the pathogenesis and longitudinal screening of IIM-ILD.

Our radiomics model, based on high-resolution chest CT texture features, achieved only moderate predictive accuracy when used by itself. This is consistent with prior studies of radiomics in ILD. Radiomics leverages quantitative analysis of CT images and captures shape, intensity, and texture patterns that are often imperceptible to the naked eye [23, 24]. In recent years, radiomics has shown promise in various ILDs. For example, CT radiomics can differentiate normal lung from fibrotic ILD and even subtypes like usual interstitial pneumonia (UIP) with high sensitivity and specificity [7, 25]. For instance, CT-based radiomic models have been developed to quantitatively predict histopathological cellular infiltration in fibrosing ILD, demonstrating the potential of radiomics for noninvasive assessment of inflammatory activity [26], and others have used radiomics to predict mortality or clinical progression in connective tissue disease [27, 28]. In systemic sclerosis (SSc)-ILD, slice-reduced CT with radiomics provided comparable diagnostic performance to full CT while lowering radiation [29]. Moreover, HRCT-based radiomic models have been shown to accurately predict rapidly progressive ILD and mortality in patients with anti-MDA5-positive dermatomyositis [30]. Thus, the radiomics approach that applies a standard U-Net lung segmentation and extracts high-dimensional features is aligned with these studies. That our radiomics-only model performed moderately suggests that, while imaging features contain signal of fibrosis, radiomics alone may not fully capture the complex systemic context of IIM. This mirrors the broader literature: radiomics can enhance ILD characterization but often achieves best results when combined with clinical or laboratory data [31, 32].

By integrating both clinical variables and CT radiomics in a joint LightGBM model, we achieved markedly improved discrimination. This synergy has been observed in other contexts. For instance, in anti-MDA5-positive dermatomyositis (a high-risk IIM subset), a combined

nomogram of clinical factors and HRCT radiomics significantly outperformed models using only imaging or only clinical data [30]. In that study the radiomics+clinical nomogram achieved AUCs of 0.877 (training) and 0.898 (testing), compared to lower AUC for the clinico-radiologic model [30]. Similarly, in a multicenter study of lateral epicondylitis treated conservatively, the authors built a LightGBM model integrating patient and MRI radiomic features; their combined model reached an external-validation AUC of 0.96, and SHAP analysis highlighted key radiographic and clinical predictors [33]. These examples underscore that fusing modalities can yield models that leverage complementary information. In our analysis, the clinical–radiomics combined model substantially outperformed the clinical-only and radiomics-only models by both statistical (higher AUC) and clinical (decision-curve) metrics. This suggests potential utility for an integrated approach in guiding screening or management: by combining symptom data, basic labs, and CT quantification, one can better stratify ILD risk than by using any single domain alone.

A key strength of this study lies in its methodological rigor in radiomics processing and modeling. Standardized preprocessing was applied, with all CT images resampled to uniform voxel spacing and gray-level intensities normalized prior to feature extraction, a step shown to mitigate scanner- or protocol-related variability [34]. A robust feature-selection pipeline was implemented to ensure the reliability of the radiomics features: reproducible features with high intra-class correlation coefficients were retained, and highly collinear variables were removed using Pearson and Spearman correlation filtering. Subsequent dimensionality reduction was performed through mRMR ranking followed by multivariable logistic regression. These steps, focusing on rigorous feature filtering and statistical parsimony, are widely recommended to minimize overfitting and ensure model stability in radiomics analyses. Lung segmentation was conducted using a publicly available U-Net (R231) model pretrained on large, heterogeneous datasets, ensuring automated and reproducible delineation of pulmonary regions [12, 35]. This standardized approach reduces manual bias and has been validated across multiple studies for accurate whole-lung segmentation.

For model interpretability, SHAP analysis was adopted to quantify each feature's contribution to individual predictions. SHAP, a game-theoretic framework, provides clear visualization of feature importance and directionality, enabling transparent understanding of how radiomic and clinical factors influence model outputs. Similar SHAP-based interpretability analyses have been increasingly applied in medical imaging to enhance trustworthiness and clinical usability of artificial intelligence models. Overall, the combination of standardized preprocessing,

multilayered feature selection, validated segmentation, and interpretable modeling represents adherence to current best practices in radiomics and AI research, strengthening the reproducibility and credibility of the study's findings.

This study has several limitations that warrant consideration. First, as a retrospective, single-center study with a relatively small sample size, the absence of external validation may limit the generalizability of our findings across different demographics or imaging protocols. Second, while the automated U-Net R231 segmentation performed reliably, it may introduce subtle inaccuracies in cases of severe or complex lung distortion, potentially affecting radiomic feature extraction. Third, due to the retrospective nature and historical recording limitations, standardized pulmonary function test (PFT) data and CT pattern classifications were not available. Although PFTs are essential for assessing physiological impairment, HRCT-based radiomics offers distinct advantages in detecting early structural abnormalities; nonetheless, future prospective cohorts should integrate PFTs to evaluate the correlation between radiomic features and lung function decline. Additionally, advanced modalities like dual-energy CT and correlations with histopathology were not explored, which could have further enhanced biological interpretability. Finally, the clinical predictors and the Rad-score identified here focus on the detection of ILD presence rather than longitudinal prognosis; thus, their ability to predict disease progression requires validation in larger prospective cohorts.

Future research should focus on external and multicenter validation, ideally through prospective longitudinal studies that assess both prevalent and incident ILD and link these early detections with long-term clinical outcomes such as respiratory decline and mortality. Expanding the model with multimodal inputs—including autoantibody profiles, serologic biomarkers (e.g., KL-6, PM/ScI, SP-D), and pulmonary function metrics—could further enhance its clinical depth. Advances in imaging, such as dual-phase CT and cine MRI, may reveal early interstitial alterations and could be integrated into future radiomic pipelines. Furthermore, while SHAP was used to enhance model transparency, future efforts should aim to develop an explainable AI-based decision support tool that can reliably identify 'asymptomatic' high-risk IIM patients, guiding timely HRCT evaluation and early intervention to ultimately improve patient survival.

Conclusions

In conclusion, the present study developed an interpretable machine learning model that integrates key clinical predictors with CT radiomics to facilitate the early identification of ILD in patients with IIM. This combined model demonstrated superior performance compared

with models based on either clinical or imaging data alone, providing a non-invasive tool for establishing the presence of ILD even in its early stages. While our findings focus on diagnostic screening rather than predicting longitudinal disease progression, this integrative approach, with further validation in larger multicenter cohorts, has the potential to enable earlier detection and timely intervention, ultimately improving clinical management for IIM patients with pulmonary involvement.

Abbreviations

AHA	Anti-histone antibody
ALB	Albumin
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
CADM	Clinically Amyopathic Dermatomyositis
CK	Creatine kinase
CK-MB	Creatine kinase-MB isoenzyme
CRP	C-reactive protein
ESR	Erythrocyte sedimentation rate
FFMI	Fat-free mass index
FFM/PMA	Fat-free mass to pectoral muscle area ratio
HB	Hemoglobin
HRCT	High-resolution computed tomography
IIF	Anti-Nuclear Antibodies
IIM	Idiopathic inflammatory myopathy
ILD	Interstitial lung disease
LDH	Lactate dehydrogenase
MDA5	Melanoma differentiation-associated gene 5
MON	Monocytes
NEU	Neutrophils
PLT	Platelets
PFT	Pulmonary function test
RP-ILD	Rapidly progressive interstitial lung disease
SAE	Splicing factor 3B subunit 1 antibody
TIF1 γ	Transcription intermediary factor 1 γ
U1-snRNP	U1 small nuclear ribonucleoprotein
UIP	Usual interstitial pneumonia
WBC	White blood cells

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13075-026-03781-2>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Acknowledgements

The authors would like to thank the clinicians and radiologists at the First Affiliated Hospital of Henan Medical University for their valuable assistance in data collection and patient management.

Authors' contributions

K.Xu conceived and supervised the study. M. Zhang, X. Zhao, B. Song, and J. Dong collected and analyzed the data. Y. Li, C. Liang, and W. Ge performed image processing and radiomics feature extraction. M. Zhang, X. Zhao, and X. Li contributed to clinical data interpretation and statistical analysis. Z. Zhang, C. Liang, and Z. Wang supervised the overall study and provided critical guidance. K. Xu, M. Zhang, X. Zhao and Z. Qin drafted the manuscript. All authors critically revised the manuscript and approved the final version for publication.

Funding

This work was supported by the Higher Education Research Project of Henan Higher Education Society (Grant No. 2025SXHLX110).

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This retrospective study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the First Affiliated Hospital of Henan Medical University. The requirement for informed consent was waived due to the retrospective design and the use of anonymized data.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form (including individual details, images, or videos).

Competing interests

The authors declare no competing interests.

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Received: 16 October 2025 / Accepted: 24 February 2026

Published online: 05 March 2026

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